Your Title

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1.2.1 Convolutions and Fourier Transforms

Multiply by the values

If we think of the n digits of each input as coefficients of a polynomial, we can calculate the product by following these steps:

1. Sample the polynomials at $2n$ points

2. Perform the pointwise multiplication of the evaluations

3. Interpolate the result to obtain the coefficients of the product

For example, to calculate $123 \cdot 45$:

$$a = 123 \longrightarrow a(x) = 1x^2 + 2x + 3$$
$$b = 45 \longrightarrow b(x) = 0x^2 + 4x + 5$$

We sample the polynomials at $2n = 6$ points:

$$\begin{aligned} a(0) &= 1 \cdot 0^2 + 2 \cdot 0 + 3 = 3 & b(0) &= 0 \cdot 0^2 + 4 \cdot 0 + 5 = 5 \\ a(1) &= 1 \cdot 1^2 + 2 \cdot 1 + 3 = 6 & b(1) &= 0 \cdot 1^2 + 4 \cdot 1 + 5 = 9 \\ a(2) &= 1 \cdot 2^2 + 2 \cdot 2 + 3 = 11 & b(2) &= 0 \cdot 2^2 + 4 \cdot 2 + 5 = 13 \\ a(3) &= 1 \cdot 3^2 + 2 \cdot 3 + 3 = 18 & b(3) &= 0 \cdot 3^2 + 4 \cdot 3 + 5 = 17 \\ a(4) &= 1 \cdot 4^2 + 2 \cdot 4 + 3 = 27 & b(4) &= 0 \cdot 4^2 + 4 \cdot 4 + 5 = 21 \\ a(5) &= 1 \cdot 5^2 + 2 \cdot 5 + 3 = 38 & b(5) &= 0 \cdot 5^2 + 4 \cdot 5 + 5 = 25 \end{aligned}$$

The pointwise multiplication of the evaluations is:

$$\begin{aligned} c(0) &= a(0) \cdot b(0) = 3 \cdot 5 = 15 \\ c(1) &= a(1) \cdot b(1) = 6 \cdot 9 = 54 \\ c(2) &= a(2) \cdot b(2) = 11 \cdot 13 = 143 \\ c(3) &= a(3) \cdot b(3) = 18 \cdot 17 = 306 \\ c(4) &= a(4) \cdot b(4) = 27 \cdot 21 = 567 \\ c(5) &= a(5) \cdot b(5) = 38 \cdot 25 = 950 \end{aligned}$$

Finally, we can interpolate the result to obtain the coefficients of the product:

$$c(x) = 4x^3 + 13x^2 + 22x + 15$$

and convert back to an integer by shifting and adding the coefficients, finding that $123 \cdot 45 = 5535$.

This approach performs the pointwise multiplication in $\mathcal{O}(n)$, but requires $\mathcal{O}(n^2)$ operations to perform the sampling step and $\mathcal{O}(n^3)$ operations to interpolate the result, making it worse than the classical method.

We are however free to choose the points at which we sample the polynomials, which can drastically reduce the amount of operations needed. Instead of sampling at $2n$ random points, we can choose the points to be the N-th roots of unity, *i.e.* we calculate the Discrete Fourier Transform of the input polynomials:

$$\mathcal{F}(x) = \sum_{n=0}^{N} x_n e^{-\frac{i2\pi}{N} kn} \tag{1.1}$$

At this point we can perform the pointwise product in the *transformed* domain and, since $\mathcal{F}$ is invertible, apply the inverse transform:

$$\mathcal{F}^{-1}(X) = \frac{1}{N} \sum_{n=0}^{N} X_n e^{i\frac{2\pi}{N} kn} \tag{1.2}$$

The algorithms that compute the discrete Fourier Transform of a sequence (or its inverse) are called fast Fourier transforms (FFT), and have an asymptotic complexity of $\mathcal{O}(n \log n)$.

1.2.2 Fast Fourier Transform

Cooley-Tukey FFT

The Cooley-Tukey FFT [**?**] is the most common FFT algorithm. It is a *divide et impera* algorithm, which recursively expresses a DFT of size $N = N_1 N_2$ into N_1 smaller DFTs of size N_2 each. The simplest form of the Cooley-Tukey algorithm is the radix-2 decimation-in-time FFT, which computes the DFTs of the even and odd indexed inputs separately and then combines the results. Assuming that N is a power of 2, we can rewrite the DFT as:

$$\begin{aligned}
X_k &= \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N} nk} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N}(2m)k} + \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N}(2m+1)k} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} mk} + e^{-\frac{2\pi i}{N} k} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} mk} \\
&= E_k + e^{-\frac{2\pi i}{N} k} O_k \qquad \text{for } k = 0, \cdots, N/2-1
\end{aligned}$$

where we split the DFT into two smaller DFTs, on the even and odd indexes respectively. Because of the periodicity of the complex exponential, we can also write:

$$\begin{aligned}
X_{k+N/2} &= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} m(k+N/2)} + e^{-\frac{2\pi i}{N}(k+N/2)} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} m(k+N/2)} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} mk} e^{-2\pi mi} + e^{-\frac{2\pi i}{N} k} e^{-\pi i} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} mk} e^{-2\pi mi} \\
&= \sum_{m=0}^{N/2-1} x_{2m} e^{-\frac{2\pi i}{N/2} mk} - e^{-\frac{2\pi i}{N} k} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-\frac{2\pi i}{N/2} mk} \\
&= E_k - e^{-\frac{2\pi i}{N} k} O_k \qquad \text{for } k = 0, \cdots, N/2-1
\end{aligned}$$

This means that we can express a DFT of size N as two DFTs of size $N/2$, where we can re-use intermediate results to calculate multiple outputs. Since the base case of the recursion has complexity $\mathcal{O}(n)$ we get the recurrence relation $T(n) = 2T(n/2) + \mathcal{O}(n)$; following the Master theorem [**?**], the computational complexity of the Cooley-Tukey FFT is $\mathcal{O}(n \log n)$.

Cantor basis

In [**?**] Cantor showed how to get a special basis in which it is possible to compute the FFT efficiently for any $m = p^n$, with p a prime number; later in [**?**] Gao

and Mateer used a simpler construction for the special case $p = 2$, where the Cantor basis $\mathbf{v} = (v_0, \cdots, v_{m-1})$ follows the structure:

$$v_0 = 1, \quad v_i^2 + v_i = v_i - 1 \text{ for } i \geq 1 \tag{1.3}$$

With respect to the Cantor basis $\mathbf{v}$, we define its sequence of subspaces as:

$$W_0 := \{0\}, \quad W_i := span\{v_0, \cdots, v_{i-1}\} \text{ for } i \geq 1 \tag{1.4}$$

By construction, we have that $W_0 \subset W_1 \subset \cdots \subset W_{m-1} = \mathbb{F}_{2^m}$ and the basis for a smaller space is contained into the basis of a larger space: this means that we can arbitrarily treat elements of smaller fields as elements in the larger fields by padding zeros to the extra dimensions.

With respect to each element of the basis v_i we can define its subspace polynomial $s_i(x)$ as:

$$s_i(x) = \prod_{a \in W_i} (x - a) \tag{1.5}$$

Since $\dim(W_i) = i$ we have $\deg(s_i(x)) = |V_i| = 2^i$. Such polynomials, as shown in [**?**, **?**], satisfy the following useful properties:

- linearity: $s_i(x + y) = s_i(x) + s_i(y)$
- recursivity: $s_{i+j}(x) = s_i(s_j(x))$

Segmentation of large polynomials

Suppose we are multiplying two binary polynomials $f(x)$ and $g(x) \in \mathbb{F}_2[x]$, represented as sequences of d bits:

$$a(x) = a_0 + a_1 x + \cdots + a_{d-1} x^{d-1}$$

In some cases, especially with high degree polynomials, it can be useful to split the polynomials into smaller segments, which can be easier to handle. The standard *Kronecker segmentation* for multiplying binary polynomials is performed as follows

1. Split the polynomials into $n = \lceil d/w \rceil$ w-bit blocks

$$\begin{aligned} a(x) &= a_0 + a_1 x + \cdots + a_{d-1} x^{d-1} \\ &= (a_0 + \cdots + a_{w-1} x^{w-1}) + \cdots + (a_{d-w} + \cdots + a_{d-1} x^{w-1}) x^{w(n-1)} \end{aligned}$$

2. Define the field $\mathbb{F}_{2^{2w}} := \mathbb{F}_2[z]/g(z)$, with $g(z)$ irreducible polynomial of degree $2w$.

3. Define the map $\psi : \mathbb{F}_2[x] \to \mathbb{F}_{2^{2w}[y]}$ such that:

$$a'(y) = \psi(a(x)) := a'_0 + a'_1 y + \cdots + a'_{n-1} y^{n-1}$$

   where $a'_i = (a_{iw} + a_{iw+1} z + \cdots + a_{iw+(w-1)} z^{w-1})$.

4. Calculate the product $c'(y) = a'(y) \cdot b'(y) = c'_0 + c'_1 y + \cdots \in \mathbb{F}_{2^{2w}}[y]$

5. Map $z \mapsto x$ and $y \mapsto x^w$, collect terms and coefficients and find the result.

At step (**??**) we employ the FFT-based method to perform the pointwise multiplication.

References